

## 02 — MOCK EXAM (questions only)

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**SQAT (CSC 4133) — Mid · Chapters 1–6 Total: 100 marks** — Section A (40 × 1) · Section B (20 × 1) · Section C (5 × 8) Time yourself: aim for ~35 min on A+B, ~55 min on C. Answers are in 03\_ANSWER\_KEY.md — don't peek until you've finished.

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### SECTION A — Multiple Choice (40 × 1 = 40 marks)

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*Choose the single best answer.*

1. Software is best defined as: a) only a computer program b) programs, procedures, documentation and data c) hardware and firmware d) source code only
2. COTS (commercial off-the-shelf) software is an example of which type? a) Customized b) Bespoke c) Generic d) Embedded
3. Which statement correctly describes SQA? a) It executes the program and detects defects b) It aims to prevent defects and does not execute the program c) Only the testing team is responsible d) It is a failure-detection system
4. The Ariane 5 rocket disaster was caused by: a) a hardware failure b) pilot error c) a control-software malfunction d) a fuel leak
5. A human action that produces an incorrect result is a(n): a) Fault b) Failure c) Error d) Defect
6. "The inability of a system or component to perform its required function" defines a: a) fault b) failure c) error d) hazard
7. Which type of analysis actually executes the program? a) Static b) Dynamic c) Inspection d) Code review
8. SQA is described as a(n) \_\_\_\_\_ activity applied throughout the software process. a) umbrella b) one-time c) post-release d) optional
9. The five views of software quality were proposed by: a) McCall b) Kitchenham & Pfleeger c) Ilene Burnstein d) ISO
10. Which view holds that quality is recognized through experience but cannot be defined? a) User b) Product c) Mystical d) Value-based
11. The CMM and ISO 9001 models are based on which view of quality? a) User b) Manufacturing c) Product d) Mystical
12. Reliability is an example of a quality: a) criterion b) factor c) metric d) standard
13. Modularity is an example of a quality: a) factor b) criterion c) view d) model
14. Which of the following is **NOT** one of the six top-level ISO-9126 characteristics? a) Functionality b) Portability c) Security d) Usability
15. "The probability of software executing without failure for a specific period of time" defines: a) Efficiency b) Usability c) Reliability d) Portability
16. The value-based view of quality merges which two concepts? a) cost and schedule b) excellence and worth c) usability and reliability d) internal and external
17. How many maturity levels does the CMM have? a) 3 b) 4 c) 5 d) 6
18. CMM Level 4 is: a) Defined b) Managed c) Optimizing d) Repeatable
19. CMM Level 5 is: a) Managed b) Optimizing c) Defined d) Initial
20. The Testing Maturity Model (TMM) was pioneered by: a) McCall b) Kitchenham c) Ilene Burnstein d) Jeff Tian
21. At which CMM level are processes "disorganized, even chaotic"? a) Initial b) Repeatable c) Defined d) Managed
22. Which model is used to **evaluate a testing** process? a) CMM b) TMM c) ISO 9001 d) McCall

23. TMM Level 3 is: a) Phase Definition b) Integration c) Management and Measurement d) Optimization
24. The three generic QA categories are prevention, reduction, and: a) detection b) containment c) correction d) validation
25. The auto-save feature of a document editor is an example of defect: a) prevention b) reduction c) containment d) removal
26. N-Version Programming (NVP) is a form of: a) defect prevention b) fault tolerance / containment c) inspection d) validation
27. A "pre-condition to an accident" is called a: a) hazard b) failure c) fault d) error
28. Which is the most commonly used **static** technique to reduce defects? a) Testing b) Simulation c) Inspection d) Prototyping
29. Which practices are highlighted for the Agile/XP process? a) waterfall stages b) test-driven development & pair programming c) big design up front d) hazard analysis
30. A test plan is a **\_\_ document, whereas a test case is a \_\_** document. a) low-level; high-level b) high-level; low-level c) macro; macro d) internal; external
31. A collection of individual test cases run in sequence until stopping criteria are met is a: a) test plan b) test case c) test suite d) test cycle
32. Reusing test cases from earlier versions of the software is called: a) unit testing b) regression testing c) smoke testing d) acceptance testing
33. A testing team organized around one product, with dedicated people, uses the \_\_\_\_\_ model. a) horizontal b) vertical c) mixed d) matrix
34. The horizontal team model: a) does all testing for one product b) performs one kind of testing for many products c) does no testing d) does only unit testing
35. In SQE, most of the key decisions about testing are made during: a) test execution b) pre-QA / planning c) post-QA d) maintenance
36. Which principle states that testing can show defects exist but cannot prove they are absent? a) Principle 1 b) Principle 2 c) Principle 5 d) Principle 7
37. The "pesticide paradox" means: a) tests are toxic b) repeating the same tests eventually finds no new defects c) defects cluster d) testing is context dependent
38. Unit testing is typically performed by the: a) end-user b) tester c) developer d) manager
39. White-box testing is also known as: a) functional testing b) structural testing c) acceptance testing d) beta testing
40. Which of the following **cannot** be fully automated? a) regression testing b) load testing c) usability testing d) data-driven testing
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## SECTION B — True / False (20 × 1 = 20 marks)

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*Write T or F.*

1. Software Quality Control (SQC) always involves executing the program. 2. In SQA, only the testing team is responsible for quality. 3. Static analysis executes the program to expose failures. 4. A fault is a human action that produces an incorrect result. 5. Security is one of the six top-level ISO-9126 quality characteristics. 6. The manufacturing view of quality is based on conformance to requirements. 7. CMM Level 5 is "Managed". 8. The TMM has five maturity levels. 9. Defect-prevention activities can be 100% effective. 10. N-Version Programming (NVP) is a defect-containment technique. 11. A test case is a high-level document. 12. Regression testing designs brand-new test cases. 13. Testing shows the presence of defects but cannot prove their absence. 14. Exhaustive testing is practical for most systems. 15. Acceptance testing is performed by the end-user. 16. White-box testing is performed mainly by developers. 17. Automated testing can fully replace manual testing. 18.

Reliability is a quality criterion, not a quality factor. **19.** The five views of software quality were proposed by Kitchenham and Pfleeger. **20.** A hazard is a failure with severe consequences.

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## SECTION C — Theory (5 × 8 = 40 marks)

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*Answer all five. Each answer must include a definition, a structure/list, and at least one example.*

**C1.** Differentiate between **SQA** and **SQC**. Give at least **five** points of difference, and state which one maps to verification and which to validation. (8)

**C2.** Describe the **five views of software quality** proposed by **Kitchenham and Pfleeger**, giving one example or note for each view. (8)

**C3.** List and explain the **five levels of the CMM**. Then briefly explain how **TMM** and **TPI** differ from CMM in purpose. (8)

**C4.** Explain the **three generic categories of QA activities** (defect prevention, defect reduction, defect containment) with an example of each. For defect prevention and defect containment, also name their **two generic techniques**. (8)

**C5.** (a) Distinguish between a **test plan**, a **test case**, and a **test suite**. (b) Compare **white-box** and **black-box** testing on at least **four** dimensions, with an example of each. (8)